

WTW 158 2017 **The Ultimate Ten Problem Differentiation Challenge**

1. Let L be any tangent line to the curve $\sqrt{x} + \sqrt{y} = \sqrt{c}$. Show that the sum of the x - and y -intercepts of L is c .
2. Find the equation of the line through the origin that is tangent to the graph of $y = \ln x$.
3. Show that the normal line to any point on the circle $x^2 + y^2 = 1$ passes through the origin.
4. Prove that if f is even then f' is odd.
5. Find all lines that are simultaneously tangent to the graph of $y = x^2 + 1$ and $y = -x^2 - 1$.
6. Find the equations of all lines through the origin that are tangent to the curve $y = x^3 - 9x^2 - 16x$.
7. Suppose that a function f is differentiable in $x = 1$ and $\lim_{h \rightarrow 0} \frac{f(1+h)}{h} = 5$. Find $f(1)$ and $f'(1)$.
8. Suppose f has the property $f(x+y) = f(x)f(y)$ for all values of x and y and that $f'(0) = f(0) = 1$. Show that f is differentiable and $f'(x) = f(x)$.
9. Given a power function $f(x) = ax^n$ with $f'(2) = 3$ and $f'(4) = 24$. Find n and a .
10. How many straight lines that pass through the origin are tangent to $y = \cos x$? Find the slopes (to one dec) of two such lines by plotting the $\cot$ graph (using software).